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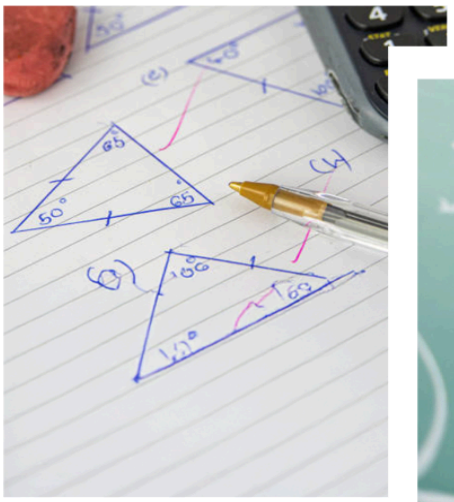
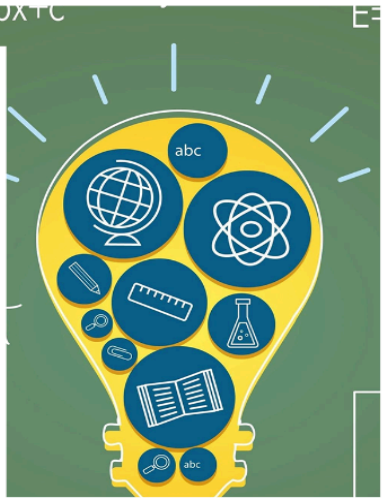
Quantitative Aptitude

Formula – Guide

Elementary Statistics





Elementary Statistics

Mean

- **Mean, $\bar{X}$** = $\frac{Some\ of\ all\ the\ observations}{Total\ no.\ of\ observations}$ = $\frac{(x_1 + x_2 + \ldots + x_n)}{n}$

- **Mean of frequency data, $\bar{X}$** = $\frac{\Sigma(fX)}{\Sigma f}$

Median

- **Median for ungrouped data**

  Arrange the numbers in ascending order i.e. listing them from smallest to largest.

  ➔ **If the number of observations are even -**

  Median = $\frac{(n/2)\ th\ term + \{(n/2)+1\}\ th\ term}{2}$

  ➔ **If the number of observations are odd -**

  Median = $\frac{(n+1)}{2}$ th term

- **Median for grouped data**

  Median = $l + \frac{(N/2 - C)}{f} \times h$

  Where, l : Lower limit of the median class

  N : Total number of observations

  C : Cumulative frequency of the class preceding the median class

  f : Frequency of the median class

  h : Size of the median class

Mode

- **Mode for ungrouped data** - The observation with the highest frequency is called a mode of data.

- **Mode for grouped data**

  Mode = $l + \frac{f1-f0}{2f1-f0-f2} \times h$

  Where, f_1: Frequency of the modal class

  f_0: Frequency of the class before the modal class

  f_2: Frequency of the class after the modal class

  h : Class width

  l : Lower class boundary of the modal class

Empirical Relationship between Mean, Median and Mode

Mode = 3 Median – 2 Mean

Range

- **Range =** Highest number - Least number

- **Coefficient of range =** $\frac{(Largest\ Item - Smallest\ Item)}{(Largest\ Item + Smallest\ Item)}$

Mean Deviation

- **Mean Deviation (MD) =** $\frac{n}{2}\ \Sigma\ |X - \bar{X}|$

  Where: X = individual value in the data set.

  $\bar{X}$ = mean value of the data set.

  n = total number of data values.

  | | = absolute value, which means it ignores the negative sign.

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Variance

- **Population Variance (σ^2) =** $\frac{1}{N} \Sigma (Xi - \mu)^2$

  Where : σ^2: Population Variance

  N : Number of observations in the population

  Xi : ith observation in the population

  μ : Population mean or Assumed mean

- **Sample variance (s^2) =** $\frac{1}{(n-1)} \Sigma (Xi - \bar{X})^2$

  Where: s^2 = Sample Variance.

  n = Number of observations in the sample.

  xi = ith observation in the sample.

  $\bar{x}$ = Sample mean or Arithmetic mean.

Standard deviation

- **Population standard deviation** is given by the formula:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (X_i - \mu)^2}$$

  Where: σ = Population standard deviation

- **The sample standard deviation** formula is as follows :

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2}$$

  Where: s = Sample standard deviation